

VIKAS BHARDWAJ

Senior AI Engineer

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 7 years of experience building and deploying production-grade AI systems across Computer Vision, Generative AI, and MLOps. Expertise in real-time video analytics, object detection and tracking, OCR, RAG, LLM fine-tuning, and model optimization. Strong hands-on experience with PyTorch, YOLO, TensorRT, Transformers, LangGraph, AWS, and GCP, with a track record of taking ML solutions from experimentation to scalable production deployment.

TECHNICAL SKILLS

Programming & Machine Learning: Python, SQL, PyTorch, TensorFlow, Scikit-learn, XGBoost, NumPy, Pandas, ML Model Development, Model Evaluation, Hyperparameter Optimization.

Agentic Frameworks Orchestration: LangChain, LangGraph, AutoGen, Multi-Agent Systems, Agent Memory, Agent Evaluation, MCP, Tool Calling, ReAct, Workflow Orchestration, Human-in-the-Loop Guardrails.

Generative AI & NLP: LLMs, RAG, LangChain, LangGraph, Hugging Face, OpenAI, Anthropic Claude, llama, Ollama, NER, DistilBERT, Transformers, Embeddings, Prompt Engineering, RAG Evaluation, LLM Evaluation Semantic Search, Reranking, LLM Fine-tuning (LoRA/QLoRA), RAG Optimization.

Computer Vision: OpenCV, Object Detection, Object Tracking, Image Classification, Vision Transformers (ViT), YOLOv5/v7/v8, OCR, Image Segmentation, BoT-SORT, TensorRT, ONNX, Model Optimization

MLOps & Cloud: AWS SageMaker, S3, ECR, Lambda, CloudWatch, GCP Vertex AI, GCS, Kubeflow, Docker, CI/CD, Model Registry, ML Pipelines, Model Versioning, Experiment Tracking, Model Monitoring, Distributed Training

Databases & Development: FastAPI, Flask, MySQL, Elasticsearch, Pinecone, Milvus, Docker, Async Processing, REST APIs, RabbitMQ, Redis, Git, GitHub

PROFESSIONAL EXPERIENCE

Pro-Vigil

Senior AI Engineer

Apr 2025 – Present

Hyderabad, India

Sentinel – Agentic Surveillance Investigation Platform

AWS Bedrock, LangGraph, Anthropic Claude

- Built a five-agent investigation workflow using **LangGraph** and **Anthropic Claude Sonnet 4** to analyze surveillance events, retrieve policies, verify evidence, and recommend incident actions.
- Developed a hybrid RAG pipeline using **OpenSearch BM25**, **MiniLM** vector search, **Reciprocal Rank Fusion**, and **Cohere reranking**, with site/time filtering and policy citation grounding.
- Processed **5M+** computer vision and monitoring records into **MongoDB**, deriving tracks, bounding boxes, OCR evidence, movement summaries, and AI-versus-track consistency checks.
- Implemented deterministic policy enforcement, claim verification, prompt-injection protection, human approval gates, evidence provenance, and auditable investigation trails.
- Created **FastAPI** and interactive CLI interfaces supporting follow-up investigations, 22 typed tools, multi-provider LLM execution, and component-level quality and latency evaluation.
- Built an evaluation framework covering RAG quality, investigation accuracy, policy leakage, hallucination, faithfulness, and latency; achieved **98% Recall@10**, **0.87 MRR**, **0.90 nDCG@10**, and **zero policy leakage**.

LlamaOps – Generative AI CI/CD & Model Deployment Platform

AWS SageMaker, CodePipeline, CloudFormation

- Built an end-to-end **AWS MLOps** pipeline for **Llama 3.2**, automating fine-tuning, evaluation, model registration, and deployment.
- Implemented automated quality, safety, latency, and approval gates using **SageMaker Model Registry**, ensuring only validated models reached production.
- Developed event-driven CI/CD workflows using **AWS CodePipeline**, **CodeBuild**, **EventBridge**, **ECR**, and **CloudFormation**.
- Deployed scalable GPU inference endpoints with prompt-response data capture and **Amazon CloudWatch** monitoring for production observability.
- Implemented endpoint auto-scaling, blue/green deployments, rollback alarms, and automated failure recovery.
- Secured **Llama 3.2** deployments across two AWS Regions using isolated **VPCs**, multi-AZ private subnets, VPC endpoints, least-privilege **IAM**, **KMS**, **Secrets Manager**, and **CloudTrail**, eliminating public model-endpoint exposure.

Jukshio Technology Innovation

Machine Learning Engineer

Apr 2021 – Feb 2025

Hyderabad, India

Customer Support Chatbot System

RAG, LangChain, OpenAI, FastAPI, AWS

- Developed a production RAG-based customer support chatbot using **LangChain**, **OpenAI**, **FastAPI**, semantic chunking, and enterprise knowledge sources.
- Improved retriever recall to **99%** through semantic chunking and increased answer precision from **80% to 95%** using a reranker model.
- Fine-tuned **DistilBERT** for NER to detect personal information in LLM responses and implemented AI guardrails for safer and more consistent outputs.
- Deployed a scalable REST API using **FastAPI**, Docker, Amazon ECR, and **AWS SageMaker**, with CloudWatch monitoring for

reliable production serving.

OCR Pipeline for JIO KYC

ASTER, XGBoost, DistilBERT

- Developed an OCR pipeline for government documents including Aadhaar, PAN Card, Voter ID, and Driving Licence using a fine-tuned **ASTER** text recognition model, achieving **97% mAP**.
- Developed an **XGBoost**-based document classification model achieving **98.5% F1-score** for accurate document categorization.
- Fine-tuned **DistilBERT** for NER to extract entities including names, dates of birth, and identification numbers, achieving **98%+ precision**.

Shopping Cart Automation System

YOLOv7, BoT-SORT, ViT, TensorRT

- Led a team of five engineers in developing an AI-powered real-time product detection, tracking, and recognition system for retail shopping carts.
- Trained **YOLOv7** for product detection, integrated **BoT-SORT** for multi-object tracking, and implemented **Vision Transformer (ViT)** based product classification.
- Optimized models using **TensorRT**, achieving more than **98% mAP** while enabling low-latency real-time inference.
- Deployed optimized computer vision models on edge devices for on-device, low-latency inference in retail environments.

Add Innovation Technology

Sep 2019 – Feb 2021

Machine Learning Engineer

Noida, India

LPG Cylinder Inspection System

OCR, Computer Vision

- Developed and optimized an OCR solution for character recognition on LPG cylinders, achieving **98.2% precision**.
- Implemented an automated cap inspection system achieving **99% precision** for identifying defective components in manufacturing workflows.

GAN-Based Defect Detection for Bike Clutch Plates

GANs, Image Segmentation

- Developed a GAN-based architecture to learn the correct structure of bike clutch plates and automatically segment defective regions.
- Enabled simultaneous classification and localization of defects for automated manufacturing inspection.

EDUCATION

Kamla Nehru Institute of Technology

Jul 2015 – May 2019

Bachelor of Technology – Information Technology

Sultanpur, India